DAX & Data Visualization

DAX Calculations:
Calculated Columns

● **Create a Calculated Column for 'Category Type':** Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

```
1 Category Type = CONCATENATE('Order Details'[Category],"-"&'Order Details'[Sub-Category])
```

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type
B-25604	65	17	2	Clothing	T-shirt	26.15%	Profit	Clothing-T-shirt
B-25606	87	4	2	Clothing	Shirt	4.60%	Profit	Clothing-Shirt
B-25607	50	15	4	Clothing	Leggings	30.00%	Profit	Clothing-Leggings
B-25615	68	20	5	Clothing	Hankerchief	29.41%	Profit	Clothing-Hankerchief
B-25616	42	12	5	Clothing	Hankerchief	28.57%	Profit	Clothing-Hankerchief
B-25619	353	90	8	Clothing	Saree	25.50%	Profit	Clothing-Saree
B-25623	53	1	4	Clothing	Stole	1.89%	Profit	Clothing-Stole

● **Calculate Revenue per Order in Order Details Table:** Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.

```
1 Revenue = ('Order Details'[Amount]*'Order Details'[Quantity])
```

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue
B-25604	65	17	2	Clothing	T-shirt	26.15%	Profit	Clothing-T-shirt	130
B-25606	87	4	2	Clothing	Shirt	4.60%	Profit	Clothing-Shirt	174
B-25607	50	15	4	Clothing	Leggings	30.00%	Profit	Clothing-Leggings	200
B-25615	68	20	5	Clothing	Hankerchief	29.41%	Profit	Clothing-Hankerchief	340
B-25616	42	12	5	Clothing	Hankerchief	28.57%	Profit	Clothing-Hankerchief	210
B-25619	353	90	8	Clothing	Saree	25.50%	Profit	Clothing-Saree	2824
B-25623	53	1	4	Clothing	Stole	1.89%	Profit	Clothing-Stole	212

● **Create a Calculated Column to Categorize Sales:** Add a calculated column named ‘Sales Category’ in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

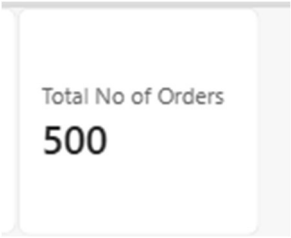
```
1 Categorize Sales = IF('Order Details'[Amount]<=270,"Below Average","Above Average")
```

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue	Categorize Sales
B-25604	65	17	2	Clothing	T-shirt	26.15%	Profit	Clothing-T-shirt	130	Below Average
B-25606	87	4	2	Clothing	Shirt	4.60%	Profit	Clothing-Shirt	174	Below Average
B-25607	50	15	4	Clothing	Leggings	30.00%	Profit	Clothing-Leggings	200	Below Average
B-25615	68	20	5	Clothing	Hankerchief	29.41%	Profit	Clothing-Hankerchief	340	Below Average
B-25616	42	12	5	Clothing	Hankerchief	28.57%	Profit	Clothing-Hankerchief	210	Below Average
B-25619	353	90	8	Clothing	Saree	25.50%	Profit	Clothing-Saree	2824	Above Average
B-25623	53	1	4	Clothing	Stole	1.89%	Profit	Clothing-Stole	212	Below Average
B-25624	26	12	3	Clothing	Hankerchief	46.15%	Profit	Clothing-Hankerchief	78	Below Average

Calculated Measures

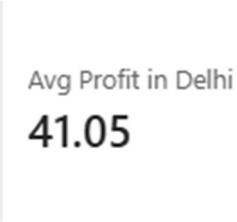
1. **Order Count-** Create a measure that calculates the **total number of orders** in the **Order Details** table.

```
Total No of Orders = COUNTA('Order Details'[Order ID])
```



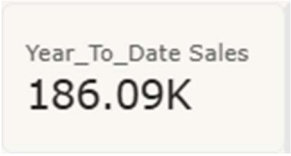
2. **Average Profit in Delhi-** Create a measure that calculates the **average profit for orders placed in Delhi**.

```
 Avg Profit in Delhi = CALCULATE(AVERAGE('Order
Details'[Profit]),FILTER('List of Orders','List of
Orders'[State]="Delhi"))
```



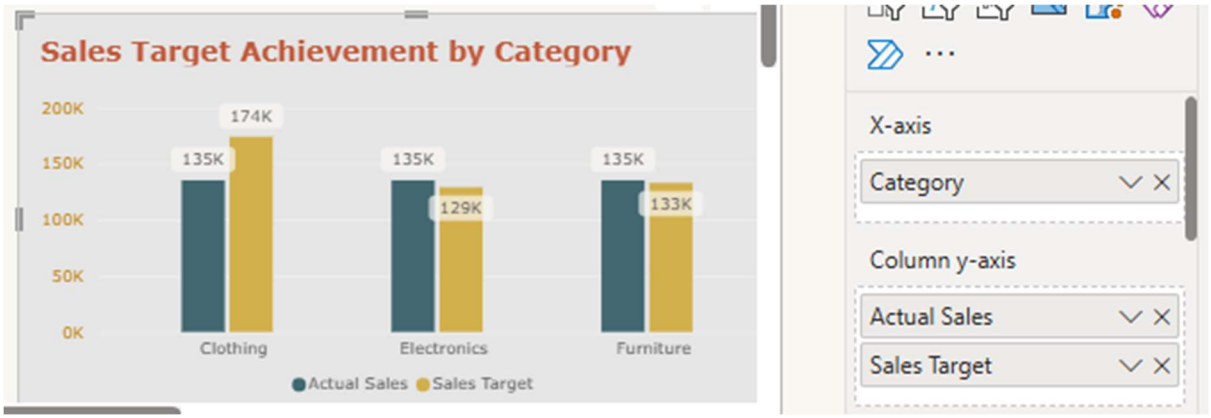
3. **Year-to-Date (YTD) Sales-** Create a measure that calculates **total sales accumulated from the beginning of the year up to the current order date**.

```
Year_To_Date Sales = TOTALYTD(sum('Order Details'[Revenue]),'List of
Orders'[Order Date].[Date])
```

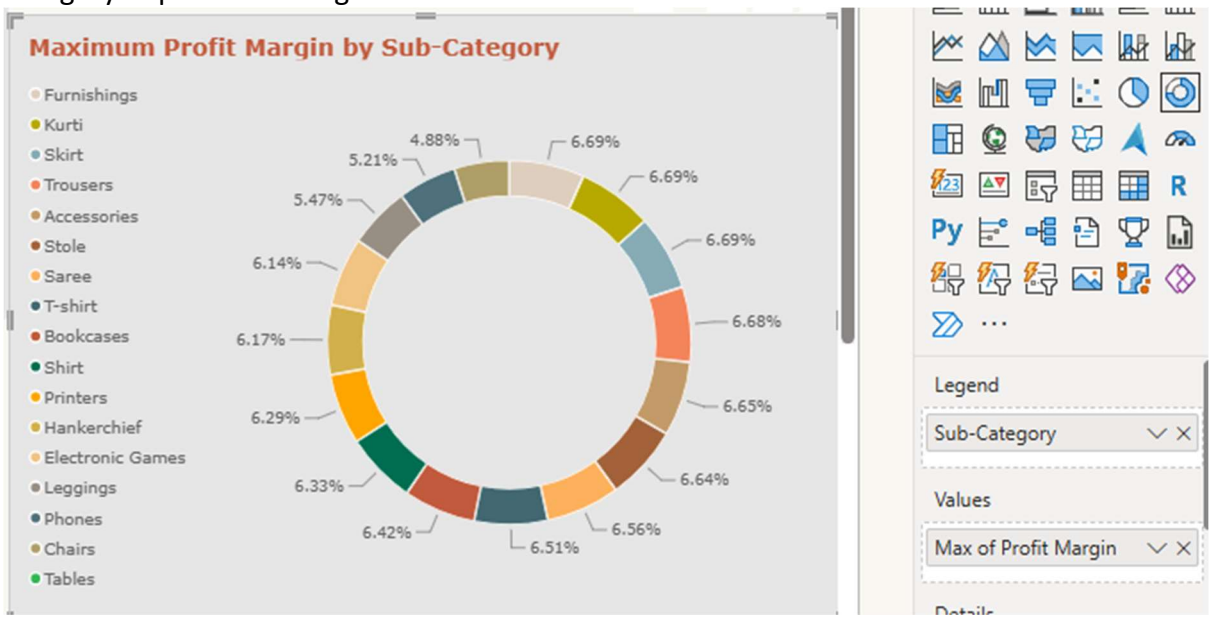


Data Visualization

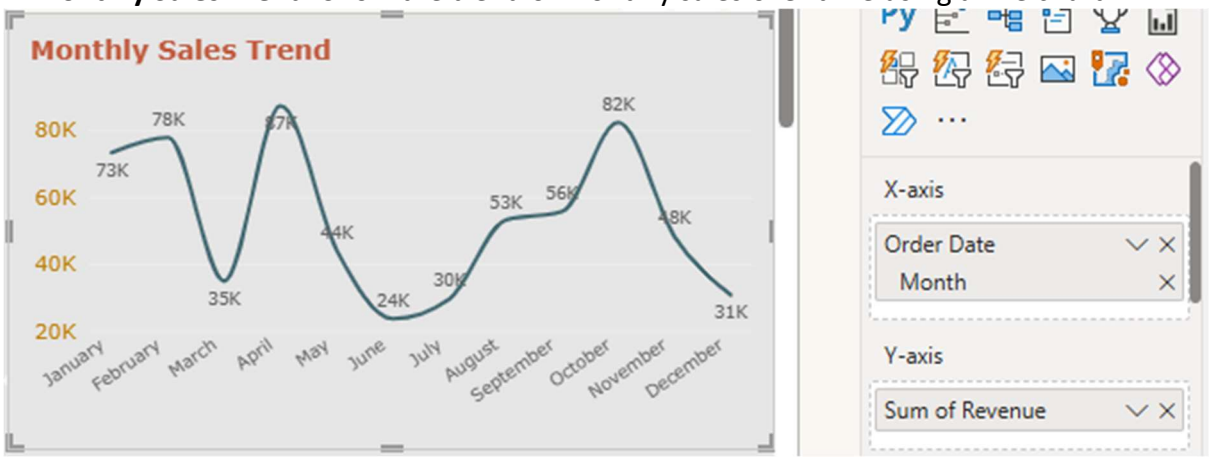
● **Sales Target Achievement by Category:** Compare actual sales with sales targets by category using a clustered column chart.



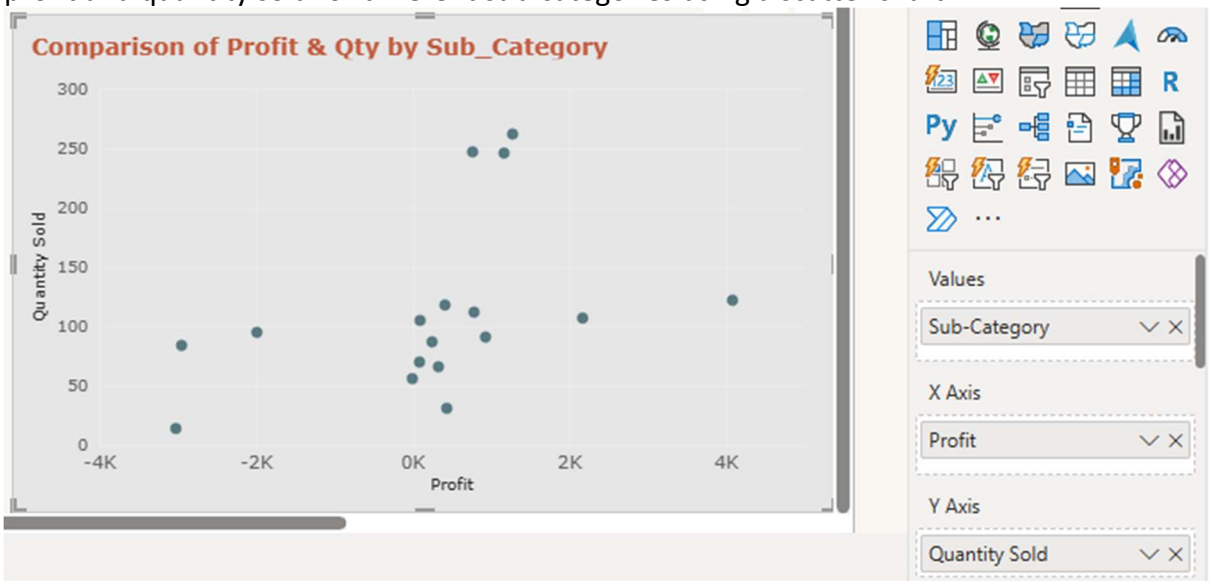
● **Max Profit Margin by Sub-Category:** Analyze the maximum profit margin for each sub-category of products using a donut chart.



● **Monthly Sales Trend:** Show the trend of monthly sales over time using a line chart.



● **Comparison of Profit and Quantity by Sub-Category:** Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.



● **Comparison of Total Sales Amount and Target:** Create cards to succinctly display the total sales amount alongside the sales target for quick comparison

